Waiting for the first success

Upper high school · probability and sequences · Progressive exercises · 08

Exercise 1. With $p = \frac{1}{4}$, find $P(T = 3)$.

Exercise 2. With $p = \frac{1}{4}$, find $P(T \leq 3)$.

Exercise 3. For $0 < p < 1$, find $P(T > m + n | T > m)$.

Exercise 4. Success has probability 0.2 at each independent trial. What is the mean waiting time? Must the fifth trial succeed?